Domain and Range

Domain from equations

1 State the domain of each function:

a $f(x) = \frac{1}{x-4}$

b $g(x) = \sqrt{2x-6}$

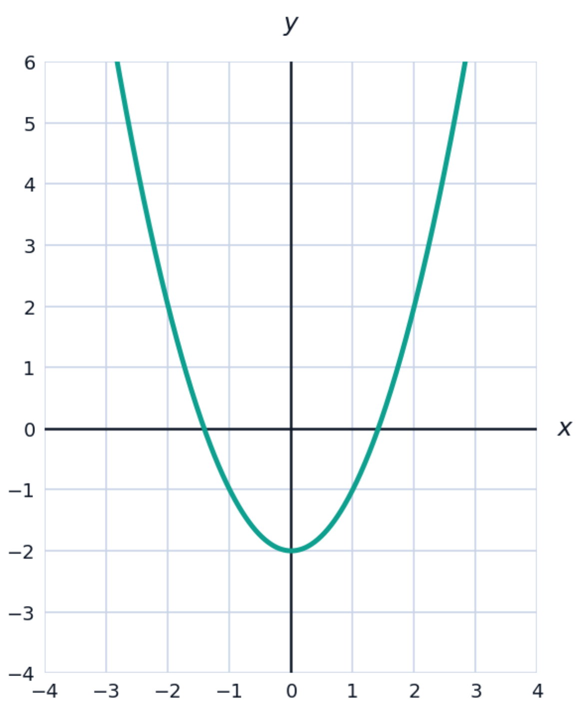
c $h(x) = \frac{x+1}{x^2-9}$

d $k(x) = \sqrt[3]{x}$

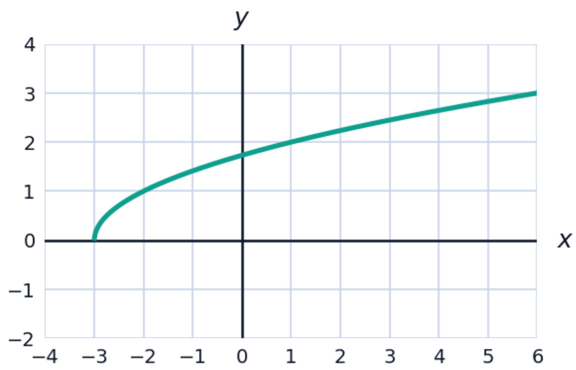
2 Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

Domain and range from graphs

3 The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.



4 The graph of $y = \sqrt{x+3}$ is shown. State the domain and range of the function.



Finding the range algebraically

5 By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

6 Consider $f(x) = -2(x-1)^2 + 5$.

a State the coordinates of the vertex.

b Does the parabola open upward or downward?

c State the range of f.